# Example 1 — Acute Decompensated HFrEF With Pneumonia and Acute Kidney Injury

**Patient:** 72-year-old man
**Admission diagnosis:** Acute hypoxic respiratory failure due to acute on chronic systolic heart failure and community-acquired pneumonia
**Discharge diagnoses:**

- Acute on chronic HFrEF, EF 30%
- Community-acquired right lower lobe pneumonia
- Acute kidney injury, improved
- Hypertension
- Type 2 diabetes mellitus
- Chronic atrial fibrillation
- Chronic anemia

**Hospital Course**

The patient presented with 5 days of progressive dyspnea, orthopnea, bilateral lower-extremity edema, and productive cough. On arrival, oxygen saturation was 84% on room air. Chest radiograph demonstrated pulmonary vascular congestion with a right lower lobe infiltrate. BNP was elevated, and echocardiogram demonstrated LVEF approximately 30%, decreased from 40% on prior study.

He was treated with IV loop diuretics with improvement in respiratory status and peripheral edema. He initially required supplemental oxygen but was weaned to room air prior to discharge. He was also treated for presumed community-acquired pneumonia with ceftriaxone and azithromycin, with clinical improvement. Blood cultures remained negative. Sputum culture grew normal respiratory flora.

Creatinine increased from 1.2 to 1.6 mg/dL during diuresis and subsequently improved to 1.3 mg/dL prior to discharge. Guideline-directed heart failure therapy was adjusted during admission. ACE inhibitor was held during the acute kidney injury and transitioned to a lower-dose ARB at discharge. Beta blocker was continued. Mineralocorticoid receptor antagonist was deferred pending repeat renal function and potassium testing.

The patient was clinically euvolemic at discharge and able to ambulate without significant dyspnea.

## Discharge Medications

### New medications

- **Furosemide 40 mg by mouth once daily** for volume management.
- **Losartan 25 mg by mouth once daily** for heart failure and blood pressure control.
- **Empagliflozin 10 mg by mouth once daily** for heart failure/diabetes management.
- **Amoxicillin-clavulanate 875/125 mg by mouth twice daily for 3 additional days** to complete treatment for pneumonia.

### Continue

- Metoprolol succinate 50 mg once daily.
- Apixaban 5 mg twice daily.
- Atorvastatin 40 mg nightly.
- Metformin 500 mg twice daily, provided renal function remains stable.

### Stopped/held

- Lisinopril — discontinued because therapy was changed to losartan.
- Spironolactone — held pending repeat renal function and potassium.
- Ibuprofen/other NSAIDs — avoid because of heart failure and recent kidney injury.

## Discharge Instructions

- Follow a **low-sodium diet**, generally targeting approximately 2 grams of sodium daily.
- Monitor daily weight at approximately the same time each morning.
- Keep a written record of weight and symptoms.
- Follow fluid recommendations provided by the heart-failure team.
- Complete the entire antibiotic course even if feeling better.
- Avoid NSAIDs unless specifically approved by the treating clinician.
- Obtain a basic metabolic panel before or at the early outpatient follow-up to reassess creatinine and potassium.
- Follow up with primary care and cardiology within approximately 1 week.

## Return Precautions

Seek urgent medical attention for:

- New or worsening shortness of breath, especially at rest.
- Chest pain or pressure.
- Fainting or severe dizziness.
- Rapidly worsening leg or abdominal swelling.
- Significant decrease in urine output.
- New confusion or severe weakness.
- Fever or worsening cough despite antibiotics.

The patient should contact the heart-failure team promptly for a **rapid increase in weight or worsening swelling/breathlessness**, according to the individualized weight-management plan provided at discharge.

**Nursing Discharge Teaching / What Nursing Typically Relays**

Nursing would typically reinforce:

1. **Medication reconciliation:** review which medications are new, continued, stopped, and temporarily held.
2. **Diuretic teaching:** explain that furosemide increases urination and discuss timing it earlier in the day when possible.
3. **Daily weights:** demonstrate how and when to weigh and what action to take if weight rises according to the physician's prescribed threshold.
4. **Heart-failure symptoms:** reinforce monitoring for increasing edema, orthopnea, shortness of breath, or rapid weight gain.
5. **Antibiotic adherence:** emphasize completing the antibiotic course.
6. **Bleeding precautions:** because the patient takes apixaban, review signs of significant bleeding and the importance of notifying clinicians about new medications.
7. Confirm that the patient understands **follow-up appointments and laboratory testing**.
8. Use teach-back: ask the patient to explain the medication changes and when they would seek medical attention.